

A DEEP LEARNING FRAMEWORK FOR FOOTBALL MATCH PREDICTION

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Abstract — Football is one of the most far reaching and famous game, along these lines anticipating the consequences of a football match represents a fascinating test. Forecast can likewise help administrators and clubs in settling on the best choices to win associations and competitions. In view of the huge financial stakes engaged with wagering, there is an interest for high prescient exactness in this industry. Rather than utilizing the customary strategies dependent on measurable examination and AI methods, this exploration paper centers around profound learning procedures that assistance in foreseeing the outcome with a high precision. Especially, LSTM model of Recurrent Neural Network is utilized in our exploration. LSTMs are a slightly modified version of RNNs that use multiplications and adds to make tiny changes to the data. Information travels through a mechanism known as cell states in LSTMs. LSTMs can selectively recall or forget things in this way. Accordingly, in this exploration paper, the result of a football coordinate is anticipated with the assistance of LSTMs and attempted to change it for better outcomes.

Keywords: Football match prediction, LSTM, RNN, English Premier League, Deep Learning.

I. INTRODUCTION

With more than many people following the activity in 188 nations, the English Premier Competition is the most watched association on earth. Once in a football fan's life, the person more likely than not wanted to know the outcomes before the game. Enormous amounts of highlights, like past group execution and match results, can be gathered in sport expectation to help different partners gauge the chances of winning or losing forthcoming matches. Due to the monetary stakes engaged with the wagering system, bookmakers, fans, and potential offers are completely keen on assessing the chances of a game early. The objective of this review is to utilize the profound learning calculation RNN with LSTM model to foresee football matches.

Anticipating the result of a coordinate is a troublesome undertaking with its own arrangement of attributes. In the past twenty years, programmed expectation of a football match result has been seriously examined, and a few scientists have introduced likelihood of match results. We proposed a profound neural organization based model to naturally anticipate the result of a football match, explicitly the English Premier League, in this review. The model is prepared on specific highlights and assessed dependent on the aftereffects of examinations. We assessed the exhibition of our proposed strategy to that of component based on the concept of

conventional AI procedures. We likewise examined the hardships and conditions wherein the proposed technique could figure a match's result.

The objective of this exploration is to give a system to precisely foreseeing football matches while keeping up with high exactness. Understanding how deep learning technology works to classify item sets requires an algorithmic study of prediction approaches.

When determining the result of a match between two groups, an individual will commonly consider factors like the groups' new exhibitions, regardless of whether the match will be played at home or away, etc. At the point when a human predicts the result of a match, factors like the human's inclination for groups, their appraisal of specific colleagues, etc. will affect the individual's decision. According to several researches, the colour of the team's uniform can impact judgments. As a result, without influencing any element, this system will assist in forecasting the match result.

A. English Premier League

With 20 groups battling for the title of English heroes, the Premier League is the high degree of English football. The Premier League is the most-watched association on earth, with one billion homes in 188 nations checking out watch the activity. It is the home of a portion of the world's most popular football clubs, players, mentors, and arenas. The association runs from August to May, with groups playing each other at home and away over the span of the period, for a sum of 380 games.

Toward the finish of the period, the last three groups in the association table are consigned to the Championship, England's second trip of football. The first and second-set groups, just as the third-put group, are supplanted by three clubs advanced from the championship toward the finish of-season end of the season games.

On the off chance that two clubs get done with similar measure of focuses, their situation in the Premier League rankings is controlled by their objective contrast, trailed by the quantity of objectives scored. Assuming the groups can't be isolated, they will be generally situated at a similar table.

Manchester United, Arsenal, Chelsea, Manchester City, Blackburn Rovers, Leicester City, and Liverpool have all won the Premier League since its creation in 1992.

A success is worth three points, a tie is for one point, and a misfortune is worth zero focuses, with the Premier League champion being the side with the most focuses toward the finish of the period. The point distribution in English premier league is given in Table 1.

Table I: Point Distribution in English Premier League

Results	Points
Win	3
Loss	0
Draw	1

B. Deep Learning

Machine learning and deep learning models can do managed learning, solo learning, and support learning. To order or conjecture, regulated learning utilizes marked datasets this requires some kind of human inclusion to properly name input information. Solo learning, then again, doesn't need named datasets all things being equal, it finds designs in the information and groups them as per any separating measures. Support learning is a technique for helping a model to expand its precision at executing an undertaking in a particular climate by utilizing input to augment the award.

Machine learning is a subset of profound realizing, which is basically a three-layer neural organization. These neural organizations attempt to mirror human cerebrum conduct, however they miss the mark concerning the mind's capacity to "learn" from huge volumes of information. While a solitary layer neural organization can just deliver approximations, adding stowed away layers can assist with further developing exactness and streamline the organization.

Deep learning is utilized in numerous man-made brainpower items and administrations to expand computerization by performing logical and actual errands without the requirement for human investment [1-3]. Deep learning innovation is utilized in ordinary items and administrations like computerized associates, voice-empowered TV controllers, and Visa extortion location, just as imminent turns of events like self-driving vehicles.

Deep learning differs from traditional machine learning in the kind of data it uses and the learning algorithms it employs.

To make predictions, machine learning calculations use organized, named information, which infers that specific properties from the information of the model are characterized and coordinated into tables. This isn't to say it never utilizes unstructured information rather, it changes it over to an organized configuration first.

Machine learning requires less data pre-processing when using deep learning. These calculations can ingest and dissect unstructured information like text and photographs, just as robotize highlight extraction, dispensing with the requirement for human trained professionals. Consider the case below: We have a collection of photos of various pets that we'd want to classify using terms like "cat", "dog", "hamster" and so on. Deep learning calculations can figure out which qualities (like ears) are generally fundamental in distinctive one creature from another. A machine learning specialist created this feature hierarchy manually.

Utilizing angle plunge and back spread, the deep learning framework changes and fits itself for exactness, permitting it

to make more exact expectations about another picture of a creature.

Thus, this research aims to predict football matches particularly, English premier league matches, using the deep learning algorithm RNN with LSTM model.

II. LITERATURE REVIEW

Hucaljuk and Rakipovid (2020) [4] utilized in-game match exercises rather than the quantity of objectives scored by each side to test a few machine learning approaches for estimating the result and aftereffect of football matches. They have tested different model architecture theories and analyse the efficiency of their models against benchmark techniques. In this paper, they have established an anticipated objective measure to help us assess the success of a team rather than use the specific achieved goals. To develop a classification model that predicts the outcomes of future matches and a regression model that predicts future matches, this measure is combined with the measurement of an offensive and defensive team ranking update during the game. The efficiency of their models correlates well with the current mainstream strategies and is close to that of bookmakers.

Baboota and Kaur (2019) [5] demonstrates their research in developing a common statistical model for the English Premier League games. They have built a feature collection using software engineering and an exploratory data review, which evaluates the main factors for predicting football match outcomes and thus develop a highly detailed predictive framework by machine learning. They have demonstrated that their model's success is highly based on important characteristics. For game weeks 6 to 38 in the EPL accumulated of two seasons (2014–2015 and 2015–2016), their best model with inclination supports accomplished an exhibition of 0.2156 in the likelihood (RPS) metric, though the wagering associations they consider (Bet365 and Pinnacle Sports) got RPS upsides of 0.2012 for a similar period. Since the low RPS esteem shows a higher prescient exactness, notwithstanding the positive outcomes, their model didn't beat the bookmaker's forecasts.

Blangiardo et al. (2020) [6] developed a Bayesian hierarchical model for football prediction. In recent years, the topic of football modelling has grown in popularity, and several different models have been developed to predict the characteristics that cause a team to lose or win a game, or to forecast a game's outcome. To meet these two objectives and check the Bayesian hierarchical model focused on data from the 1991-1992 Italian Series A championship. They recommend a more complex blend model that matches the observed data to solve the over-reduction problem generated by the Bayesian hierarchical model. The Italian Series A championship 2007-2008 is an illustration for checking its results.

Zaveri et al. 2018 [7-8] made an machine learning based framework that can cover all of the current requirements in football match expectation. The execution just incorporates groups from the Spanish La Liga for the last five seasons. The last score, beginning 11 players, substitutions, and likely objective scorers have all been anticipated for matches including the Home Team and the Away Team. For investigation, we've joined player and group details from the FIFA 18 game information base, just as their genuine profession details. Individual players and groups' qualities, blemishes, and strategies have likewise been analyzed. At long last, we'd plan a framework that can evaluate both the home and away groups and afterward offer methodologies to the client for their group to assist them with amplifying their exhibition.

In view of true measurements information from the 2006 World Cup Football Game, a multi-facet perceptron (MLP) utilizing back-engendering learning strategy is utilized to expect the triumph paces of two groups in (Huang and Chen 2011) [9]. There are three sorts of preparing tests: win, draw, and rout. In the new stage, new preparing tests are gathered from past stages and added to the current preparing tests, following which the neural organization is fabricated. They had the option to accomplish a 75 percent precision in this model by barring games that completed in a tie. The match expectation works introduced in this exploration are summed up. The above research supports the appreciation and utility of the task.

The purpose of this study, according to Khazaal et al. (2015), [10] was to see if football specialists could predict football match scores better than non-experts. The precision of football match prognoses does not seem to be affected by experience, age, and sex. The assumption that soccer expertise enhances betting skills is, therefore, nothing more than a cognitive delusion known as the “illusion of control” gamblers may profit from psychological therapies that work on the illusion of control ties that their perceptions have between betting skills and soccer expertise. The training that should be considered is the public security plan to debilitate football betting.

Kampakis et al. (2014) [11] investigated to determine how Twitter data can be used. They made a model dependent on tweets and tried on the off chance that these models can beat expectation models dependent on verifiable information and straightforward soccer insights in anticipating the result of English Premier League football matches. Moreover, both Twitter and historical data are used to construct combined models. When measured by the Cohen Kappa, the final Twitter-based model outperforms chance and is equivalent to a model based on simple statistics and historical data. The convergence of the two models increases performance better than the other model. This work also shows that technologies obtained from Twitter can theoretically provide valuable information to predict soccer outcomes.

Chao-Ying Joanne Peng et al. (2002) [12] showed the favored example for utilizing strategic techniques with an illustration of the calculated relapse utilized for an informational index to test an exploration speculation. Sensible announcing arrangements and least perception to-indicator proportion for calculated relapse tests are additionally proposed. In eight papers delivered in The Journal of Educational Research somewhere in the range of 1990 and 2000, the researchers talked about the use of and comprehension of useful relapse. Each of the 8 reports additionally shown that the endorsed prerequisites were satisfied or outperformed.

Prasetio and Harlili (2016) [13] made the article that developed from 2010-2011 upto 2015-2016 using variations of training data. Logistic regression is a classification tool used to forecast sports outcomes that may offer empirical insight by coefficients of regression. The criteria used are “HomeOffensive”, “HomeDefense”, “Forgotten”, and By modifying the training data used, they performed experiments. The statistical precision of the model being developed is 69.5%. They arrived at the conclusion that “Home Defense” and “Away Security” were the relevant variables.

In Bunker and Thabtah 2017, the authors developed a model that tracks a group's hostile and cautious abilities. Based

on the quantity of anticipated objectives, their model beat Dixon and Coles' (1997) model on matches from the 2013/2014 and 2014/2015 English Premier League seasons (Dixon and Coles 1997).

O'Donoghue et al. (2004) [14] utilized an assortment of strategies to foresee the results of the 2002 FIFA World Cup, including probabilistic neural organizations, straight and calculated relapse, bookmakers' chances, virtual experiences, and master conjectures. There have additionally been reads up into evaluating frameworks for anticipating the result of football match-ups. Which rating framework in (Elo 1978) is the most notable technique proposed by ArphadElo for chess game forecast and afterward applied to football (Hvattum and Arntzen 2010).

Tax and Jousts (2015) [15] made a framework for the Dutch Eredivisie focused on public data is defined in this research. A systematic literature review described the variables of predictive utility for the match outcomes. Candidate characteristics have been created. Self-made public data collection, consisting of 13 Dutch Eredivisie match data seasons, was accompanied by modelling preparation. A variety of variations have been evaluated on public data testing developed in the dimensional reduction techniques and classification algorithms. A mixture of PCA (with a difference of 15 percent) and a Naive Bayes or Multilayer Perceptron classifier obtained the best detection precision for the public data feature collection. Models for betting odds have been developed, as well as a hybrid feature set (common data union and wagering odds characteristics). Check McNemar has found no substantial gap in the accuracy of the model with the lowest accuracy hybrid function setting and the low precision betting odds, but the findings do lift the supposition that a mixed combination of betting odds and public data will defeat the bookmaker. The acquired results can be seen as a sign that competitive frameworks for supporting betting decisions based on open data can be developed.

Ancona et al. (2001) [16] have tackled the question of achieving a target during a football match utilizing strategies and devices that neither alters the ball nor the goalmouth. Automatic goal identification in football is a popular problem and the referee associations, the press and fans pay special attention to it. Sometimes, the official and the workers (linemen) are not able to discern the goal or, what is more significant, they do not assign the award accurately to either side. There are several unusual cases in which the goal happens. The occurrence happens by detecting the ball and measuring its place on the location of the postage in the photograph. The strategy for ball detection is based on a controlled classification method named vector aids. Examples are used for training the correctly filtered and previously recorded picture pattern variant of the views of the entity to be detected. On actual photographs in which the ball is both fully transparent and partly occluded, they have checked thoroughly the methodology. Detection rate, false-positive and precise ball location in the image measure performance of the proposed detection scheme.

Elfrink et al. 2018) [17] explored if it is possible to forecast which team will win individual MLB games. To acquire the consequences of their examination, they utilized chronicled information from games and a few machine learning techniques like arbitrary timberlands and XGBoost. The XGBoost calculation delivered the best outcomes, with an accuracy of 0:5552. This result, as indicated by the creator, can be improved by utilizing more information, PC power, and better element designing.

Liu et al. (2017) [18] research three straightforward models and use the DOE (Design of Experiment) technique to analyze their impact on sports forecast. Another goal is to explore the meaning of the three parts. The NBA (National Basketball Association) game

information is utilized for this examination information displaying. All of the data was assembled from an authority and openly accessible information hotspot for the normal period of 2015-2016.

III. PROPOSED SYSTEM

We will utilize the LSTM model of intermittent neural organizations in the proposed model to conjecture the consequences of the English Premier League.

Recurrent Neural Networks have an ability to simulate remembering older instances of data. Hence, it makes sense to use it in applications like Natural Language Processing. In the previous mentioned application, if a sentence is given as an instance, it is pretty obvious that the model should make sense of the sentence in its original order. This can be achieved by feeding the model, word by word, taking advantage of the properties of RNN. But the RNN cannot remember the information for the long period for time. As a result, this study is based on the Recurrent Neural Network's Long Short Term Memory.

The vital objective of this examination is to keep up with track of the group's advancement and to apply the right calculations to work on the model's general presentation. With profound learning comes a crowd of benefits like its exceptionally versatile nature, this implies that the exhibition of profound learning calculations don't level after a specific measure of information has been prepared not normal for customary AI calculations and its famous capacity in managed gaining from marked information. Profound learning calculations, which are basically profound learning organizations, endeavor to use the obscure design in the info appropriation to reveal great portrayals, which are normally at many levels, with more significant level learnt highlights indicated as far as lower-level attributes.

A. Proposed System Modules

The modules of the proposed system are dataset collection, preprocessing, training the model, model validation and prediction of the model.

B. Dataset Description

The dataset for foreseeing the football match champ was gotten from the data.uk web source (<http://football-data.co.uk/>), where we got information from the **English Premier League from seasons 2009 to 2015**. The general Number of matches played in the competition, just as the quantity of matches played by each group, are both foreordained when utilizing these football datasets. This helped us in preprocessing the data and removing the unnecessary information. Each of the league datasets have 380 rows. In addition, the form of each match's rivals was computed by looking up the **previous five matches' win streaks**, and additional attributes were created to track if any of the teams had a hat trick or had won five matches in a row. The attributes of the dataset are presented in Table II. Finally, for classification purposes, One Hot Encoding was utilized for the characterization reason, for example the match result.

C. Data Pre-processing

Batch normalization technique is applied to normalize the values in the dataset. Batch normalisation is a network input standardisation approach that can be performed to either prior layer activations or the inputs themselves. Batch normalisation reduces generalisation error by speeding up training (in some

situations by halving or bettering the epochs) and providing some regularisation.

D. Long short Term Memory

Long Short-Term Memory (LSTM) cells are a development over RNNs. In RNN, the output of every state is directly sent as the $ht-1$ to calculate the value of the new state. LSTMs are cells that implement RNNs as well as execute computations with the previous output and current input to generate the new output. Three gates make up the majority of LSTM cells. Figure 1 shows the Gate structure of LSTM.

- **Forget Gate** - When switching to a new sequence, this is responsible for forgetting the useless information. A sigmoid function is used in this gate.
- **Input Gate** - This is in charge of informing the cell about fresh information. A tanh function and a sigmoid function are used in this gate.
- **Output Gate** - The ultimate output from the cell state calculated in the preceding two states is shaped by this. The state's final output is also computed using a tanh function and a sigmoid function.

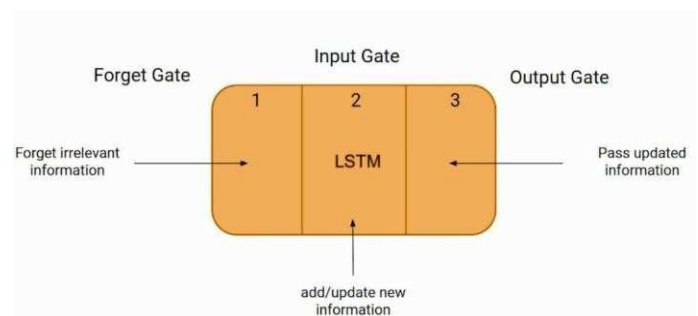


Figure 1. Gate structure of LSTM

Table II: Attributes in the dataset

Attributes	Expansion	Description
HTGS	Home Team Goals Scored	The number of goals the home team has scored.
ATGS	Away Team Goals Scored	The number of goals the away team has scored.
HTGC	Home Team Goals Conceded	The number of goals the home team has conceded.
ATGC	Away Team Goals conceded	The number of goals the away team has conceded.
HTP	Home Team Points	Represents the points scored by home team
ATP	Away Team Points	Represents the points scored by away team
HTGD	Home Team Goal Difference	Goal differences of the home team from the away team
ATGD	Away Team Goal Difference	Goal differences of the away team from the home team

HM	Home Match	Matches played in the home ground
AM	Away Match	Matches played in the away ground
HtWin Streak3	Home Team Consecutive Win(3)	Represents the number of times Home team had won three consecutive matches
AtWin Streak3	Away Team Consecutive Win(3)	Represents the number of times Away team had won three consecutive matches
HtWinStreak 5	Home Team Consecutive Win(5)	Represents the number of times Home team had won five consecutive matches.
AtWinStreak 5	Away Team Consecutive Win(5)	Represents the number of times Away team had won five consecutive matches.
HtLoss Streak3	Home Team Consecutive Lose(3)	Represents the number of times Home team had lost three consecutive matches
AtLoss Streak3	Away Team Consecutive Lose(3)	Represents the number of times Away team has lost three consecutive matches
HtLoss Streak5	Home Team Consecutive Lose(5))	Represents the number of times Home team had lost five consecutive matches.
AtLoss Streak5	Away Team Consecutive Lose(5)	Represents the number of times Away team had lost five consecutive matches.

E. Performance Measures

Interpretation gives us the ability to question the model's decision and learn about the following aspects. The proposed system uses the following measures for the evaluation of the models implemented

1. Confusion Matrix

Looking at the confusion matrix is a much better technique to assess a classifier's performance. The basic concept is to keep track of how many times examples of class A are categorized as class B. A confusion matrix has each row representing an actual class and each column representing a forecasted class. Although the confusion matrix provides a lot of information, some people prefer a shorter metric.

TP (True Positive) - when a case was positive and predicted positive

TN (True Negative) - when a case was negative and predicted negative.

FN (False Negative) - when a case was positive but predicted negative.

FP (False Positive) - when a case was negative but predicted positive.

2. Classification Report

The following are the prediction measures that are used for model evaluation described in the proposed system,

a) Precision

Precision is the capacity of a classifier to try not to mark a negative event as certain. The proportion of genuine up-sides of the amount of genuine up-sides and bogus up-sides for each class is utilized to figure it.

Precision :- Accuracy of positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

The model's ability to forecast the instance positive that is truly positive is given by equation (1)above.

b) Recall

Recall is a term that depicts a classifier's capacity to track down every certain model. The proportion of genuine up-sides of the amount of genuine up-sides and bogus negatives for each class is utilized to process it.

$$\text{Recall} = \frac{TP}{TP + FN}$$

The model's capacity to predict all positive instances is Given by equation (2)

c) F1 Score

With 1.0 being the highest and 0.0 being the lowest, the F1 score is a weighted harmonic mean of precision and recall. Because precision and recall are taken into account, F1 ratings are lower than accuracy assessments. To compare classifier models, utilize the weighted average of F1 rather than global accuracy as a rule of thumb.

$$\text{F1 Score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

The above equation (3) measures the accuracy with weighted average using the precision and recall measures.

d) Support

The quantity of genuine events of the class in the gave dataset is known as help. Imbalanced help in the preparation information could uncover essential issues in the classifier's accounted for scores, requiring delineated examining or rebalancing. Support doesn't change contingent upon the model, yet rather analyze the assessment interaction.

e) Macro Average

It's a function that computes measures for each label in the dataset and returns the average without taking the proportion into account.

f) Weighted Average:

It's a function that computes measurements for each label and delivers an average based on the proportion of labels in the dataset.

IV. RESULTS AND DISCUSSION

The viability of our proposed technique is estimated dependent on the exploratory outcomes accomplished. Figure 2 shows the characterization report of the proposed model.

Experiments are performed in English premier league datasets using the LSTM model of the recurrent neural networks. In which the LSTM model gives the accuracy of 86%.

```
print(classification_report(y_test, y_pred))
# 0 - Win, 1 - Loss, 2 - Draw
```

	precision	recall	f1-score	support
0	0.88	0.95	0.92	230
1	0.85	0.84	0.85	366
2	0.85	0.78	0.81	204
accuracy			0.86	800
macro avg	0.86	0.86	0.86	800
weighted avg	0.86	0.86	0.86	800

Figure 2. Classification Report of LSTM

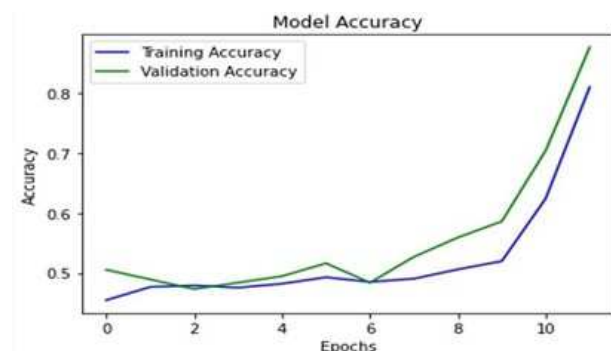


Figure 3. Graph for Model accuracy

The model accuracy graph in Figure 3 depicts how well the model's accuracy can be improved over time using

increasing epochs. It is observed, when the epochs are in the range of 0 to 8, the accuracy grows gradually but in the range of 50 % to 60 %, and when the epochs are more than 8, the accuracy increases suddenly upto 80 % to 90%.

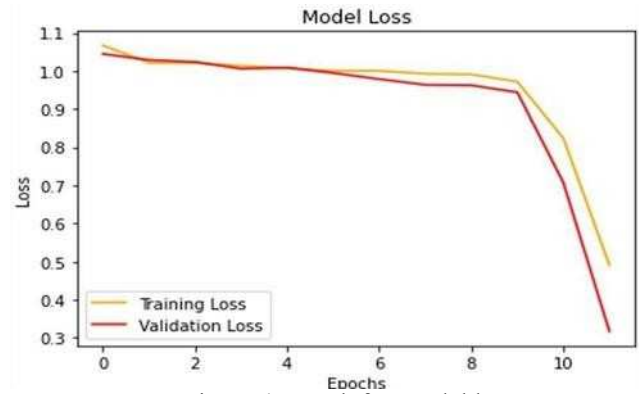


Figure 4. Graph for Model loss

The model loss graph in Figure 4 depicts how successfully the model's loss can be reduced over time using increasing epochs. It is observed that when the epochs are in the range of 0 to 8, the loss drops gradually in the range 1.1 to 0.9, and when the epochs are more than 8, the loss decreases suddenly towards 0.3.

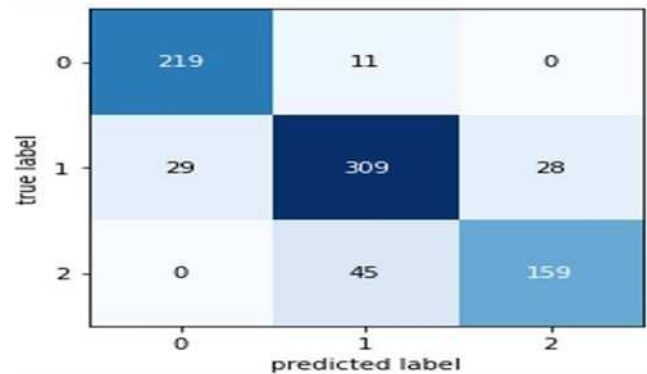


Figure 5. Confusion Matrix

The amount of true labels with respect to predicted labels for win, defeat, and draw are shown in the confusion matrix. Here, '0' denotes a win, '1' denotes a loss, and '2' denotes a draw.

V. CONCLUSION AND FUTURE WORK

Football's popularity and international impact make it an intriguing challenge to solve. Furthermore, there are more factors that influence the outcome of a match. Our proposed study employs an LSTM model to forecast whether English Premier League matches will end in a victory, defeat, or tie. According to the findings, RNNs with LSTM have a clear and obvious advantage over traditional machine learning algorithms and ANNs.

Other things outside the winner can be predicted, such as the amount of goals or cards a team will receive, as well as the minute in which they will score, using various datasets, inputs, and architectures. We can also predict the match even with high accuracy by training with the datasets having weather reports of the particular location where the match is going to held and using the social media conversations etc.

This model has room for improvement. One approach to do so is to use a larger number of datasets. They can include individual player statistics. This can like wise help with extending a player's structure from one season to another. Gigantic datasets will like wise help with the neural organization's preparation.

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